directions of jets. [Ch. 9]

Chandrasekhar embarks on a five-year quest to develop a complete mathematical description of perturbations of black holes. [Ch. 7]

Unruh and Davies infer that, as seen by accelerating observers just above a black hole's horizon, the hole is surrounded by a hot atmosphere of particles, whose gradual escape accounts for the hole's evaporation. [Ch. 12]

Page computes the spectrum of particles radiated by black holes. Hawking and Page, from observational data on cosmic gamma rays, infer that there can be no more than 300 tiny, primordial, evaporating black holes in each cubic light-year of space. [Ch. 12]

The golden age of theoretical black-hole research is declared finished by youthful researchers. [Ch. 7]

1977 Gibbons and Hawking verify Bekenstein's conjecture that a black hole's entropy is the logarithm of the number of ways it might have been made. [Ch. 12]

Radio astronomers use interferometers to discover the jets that feed power from a galaxy's central black-hole engine to its giant radio-emitting lobes. [Ch. 9]

Blandford and Znajek show that magnetic fields, threading the horizon of a spinning black hole, can extract the hole's spin energy, and that the extracted energy can power quasars and radio galaxies. [Ch. 9]

Znajek and Damour formulate the membrane description of a black-hole horizon. [Ch. 11]

Braginsky and colleagues, and Caves, Thorne, and colleagues, devise quantum nondemolition sensors for circumventing the quantum limit on bar detectors of gravitational waves. [Ch. 10]

1978 Giacconi's group completes construction of the first high-